



graphkit – Neural Solver Development Library

Development Framework

Data Loading

- Task-specific transforms
- Data normalization
- Data splitting



datakit Datasets

Modular Components

- Command-line interface
- Model registry
- Training & evaluation
- MLFlow tracking
- Model compilation



Supported Tasks

Power Flow

Optimal Power Flow

State Estimation

GridFM-PF v0.1 & v0.2

GRIT-PF

GENCO-PF|OPF|SE



- Iterative correction steps
- Constraint-handling modules
- Task-based physics decoder
- Base, small & tiny versions

Grid Representation

- PGLib library
- Matpower files (.m)
- Topology & base load
- Grids up to 10k/30k (OPF/PF)
- Generator & bus nodes



Perturbation Pipeline

- Global load scaling + local noise
- Topology & admittance scaling
- Generator cost-coeff. permutations
- Off-nominal operating scenarios
- AC & DC runtime logging



Modes

PF

OPF

AC

DC

Datasets



- datakit
- PFΔ
- OPFData

Hosted by:

OLFENERGY
OpenGridFM



datakit – Synthetic Data Generation Library